ST. PETERSBURG STATE UNIVERSITY
Faculty of Mathematics and Mechanics

L/R No. 4: Parallel.

Made by Lichkovakh Daniil Andreevich 24 B 81-mm.

Teachers:
Vyacheslav Grokhovsky
Victor Katz
Andrey Isakov

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Figure 1 shows a clearly improved performance.
For the first image, the clockwise rotation was accelerated by 2 ms, counterclockwise by 3 ms. The application of the blurr was accelerated by 40 ms, that is, it is now done 3 times faster.
For the second image, the clockwise rotation was accelerated by 17 ms, counterclockwise by 16 ms. The application of the blurr was accelerated by 139 ms.

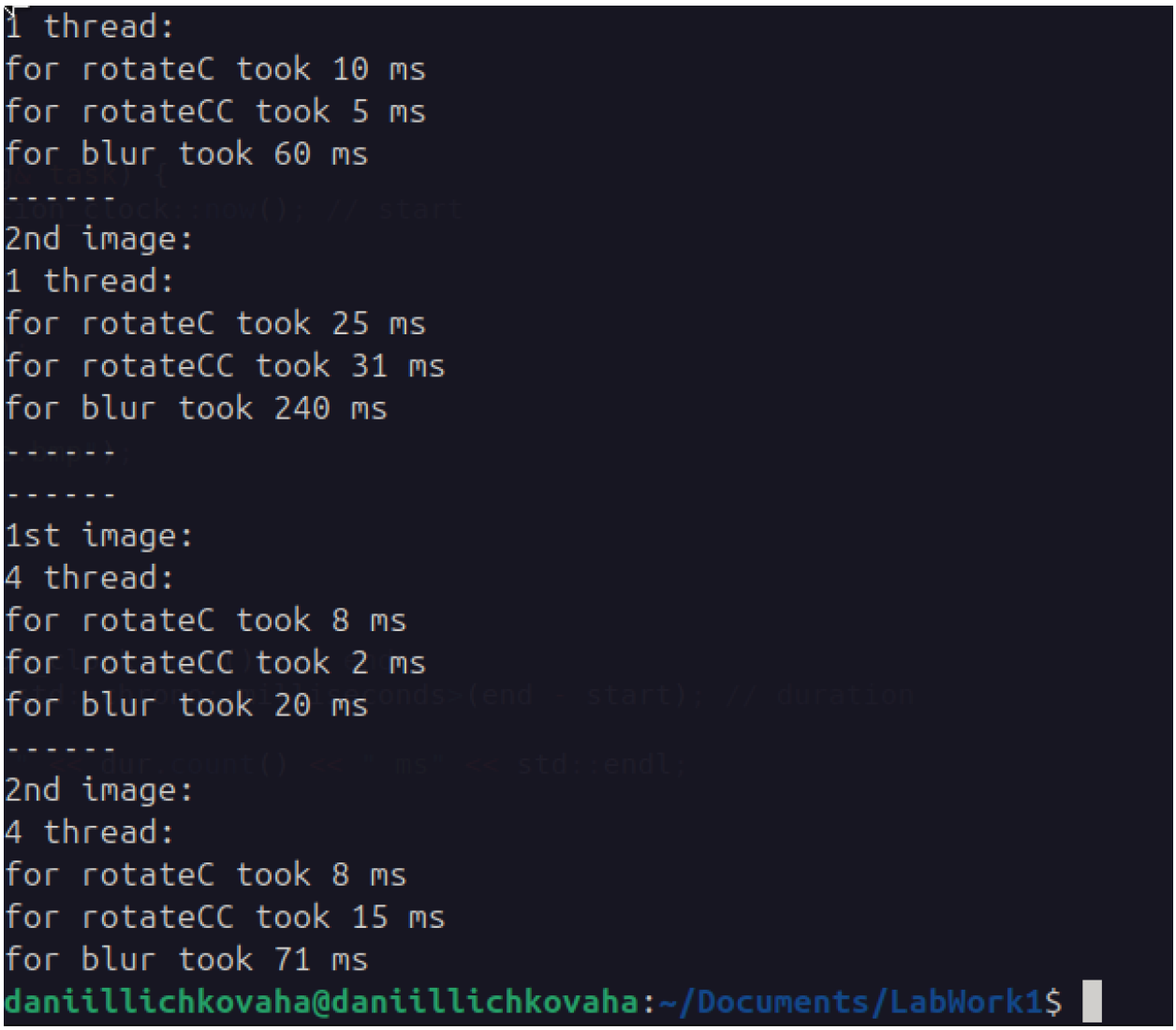

Fig. 1 - the result of the acceleration of the program.

Explanation of the drawing:
"rotateC" is a rotate clockwise
"rotateCC" is a rotate counter clockwise
blur is Gaussian Blur

It can be stated that parallelization has significantly reduced the execution time of the program. It is worth noting that the tests were carried out thanks to the testParal function with the following implementation:

```cpp
#include "BMP.h"
#include "testParal.h"
#include <iostream>
#include <string>
#include <chrono>

void testParal(BMP& img, const std::string& task) {
    auto start = std::chrono::high_resolution_clock::now(); // start

    if (task == "rotateC") {
        img.Rotate90Clockwise();
        img.Save("Rotated90Clockwise.bmp");
    } else if (task == "rotateCC") {
        img.Rotate90CounterClockwise();
        img.Save("Rotate90CounterClockwise.bmp");
    } else if (task == "blur") {
        img.GaussianBlur();
        img.Save("GaussianBlur.bmp");
    }

    auto end = std::chrono::high_resolution_clock::now(); // end
    auto dur = std::chrono::duration_cast<std::chrono::milliseconds>(end - start); // duration

    std::cout << "for " << task << " took " << dur.count() << " ms" << std::endl;
}
```

Fig. 2 - implementation of the testParal function.